

DERIVATIONS

1. Complete the solution of the brachistochrone problem begun in Section 2.2 and show that the desired curve is a cycloid with a cusp at the initial point at which the particle is released. Show also that if the particle is projected with an initial kinetic energy $\frac{1}{2}mv_0^2$ that the brachistochrone is still a cycloid passing through the two points with a cusp at a height z above the initial point given by $v_0^2 = 2gz$.
2. Show that if the potential in the Lagrangian contains velocity-dependent terms, the canonical momentum corresponding to a coordinate of rotation θ of the entire system is no longer the mechanical angular momentum L_θ but is given by

$$p_\theta = L_\theta - \sum_i \mathbf{n} \cdot \mathbf{r}_i \times \nabla_{\mathbf{v}_i} U,$$

where $\nabla_{\mathbf{v}}$ is the gradient operator in which the derivatives are with respect to the velocity components and $\mathbf{n}$ is a unit vector in the direction of rotation. If the forces are electromagnetic in character, the canonical momentum is therefore

$$p_\theta = L_\theta + \sum_i \mathbf{n} \cdot \mathbf{r}_i \times \frac{q_i}{c} \mathbf{A}_i.$$

3. Prove that the shortest distance between two points in space is a straight line.
4. Show that the geodesics of a spherical surface are great circles, i.e., circles whose centers lie at the center of the sphere.

EXERCISES

5. A particle is subjected to the potential $V(x) = -Fx$, where F is a constant. The particle travels from $x = 0$ to $x = a$ in a time interval t_0 . Assume the motion of the particle can be expressed in the form $x(t) = A + Bt + Ct^2$. Find the values of A , B , and C such that the action is a minimum.
6. Find the Euler–Lagrange equation describing the brachistochrone curve for a particle moving *inside* a spherical Earth of uniform mass density. Obtain a first integral for this differential equation by analogy to the Jacobi integral h . With the help of this integral, show that the desired curve is a hypocycloid (the curve described by a point on a circle rolling on the inside of a larger circle). Obtain an expression for the time of travel along the brachistochrone between two points on Earth's surface. How long would it take to go from New York to Los Angeles (assumed to be 4800 km apart on the surface) along a brachistochrone tunnel (assuming no friction) and how far below the surface would the deepest point of the tunnel be?
7. In Example 2 of Section 2.1 we considered the problem of the minimum surface of revolution. Examine the symmetric case $x_1 = x_2$, $y_2 = -y_1 > 0$, and express the condition for the parameter a as a transcendental equation in terms of the dimensionless quantities $k = x_2/a$, and $\alpha = y_2/x_2$. Show that for α greater than a certain value α_0 two values of k are possible, for $\alpha = \alpha_0$ only one value of k is possible, while if $\alpha < \alpha_0$ no real value of k (or a) can be found, so that no catenary solution exists in this region. Find the value of α_0 , numerically if necessary.

8. The broken-segment solution described in the text (cf. p. 42), in which the area of revolution is only that of the end circles of radius y_1 and y_2 , respectively, is known as the *Goldschmidt solution*. For the symmetric situation discussed in Exercise 7, obtain an expression for the ratio of the area generated by the catenary solutions to that given by the Goldschmidt solution. Your result should be a function only of the parameters k and α . Show that for sufficiently large values of α at least one of the catenaries gives an area below that of the Goldschmidt solution. On the other hand, show that if $\alpha = \alpha_0$, the Goldschmidt solution gives a lower area than the catenary.
9. A chain or rope of indefinite length passes freely over pulleys at heights y_1 and y_2 above the plane surface of Earth, with a horizontal distance $x_2 - x_1$ between them. If the chain or rope has a uniform linear mass density, show that the problem of finding the curve assumed between the pulleys is identical with that of the problem of minimum surface of revolution. (The transition to the Goldschmidt solution as the heights y_1 and y_2 are changed makes for a striking lecture demonstration. See Exercise 8.)
10. Suppose it is known experimentally that a particle fell a given distance y_0 in a time $t_0 = \sqrt{2y_0/g}$. The times of fall for distances other than y_0 are not known. Suppose further that the Lagrangian for the problem is known, but that instead of solving the equation of motion for y as a function of t , it is guessed that the functional form is

$$y = at + bt^2.$$

If the constants a and b are adjusted always so that the time to fall y_0 is correctly given by t_0 , show directly that the integral

$$\int_0^{t_0} L dt$$

is an extremum for real values of the coefficients only when $a = 0$ and $b = g/2$.

11. When two billiard balls collide, the instantaneous forces between them are very large but act only in an infinitesimal time Δt , in such a manner that the quantity

$$\int_{\Delta t} F dt$$

remains finite. Such forces are described as *impulsive* forces, and the integral over Δt is known as the *impulse* of the force. Show that if impulsive forces are present Lagrange's equations may be transformed into

$$\left(\frac{\partial L}{\partial \dot{q}_j} \right)_f - \left(\frac{\partial L}{\partial \dot{q}_j} \right)_i = S_j,$$

where the subscripts i and f refer to the state of the system before and after the impulse, S_j is the impulse of the generalized impulsive force corresponding to q_j , and L is the Lagrangian including all the nonimpulsive forces.

12. The term *generalized mechanics* has come to designate a variety of classical mechanics in which the Lagrangian contains time derivatives of q_i higher than the first. Problems for which $\ddot{x} = f(x, \dot{x}, \ddot{x}, t)$ have been referred to as “jerky” mechanics. Such

equations of motion have interesting applications in chaos theory (cf. Chapter 11). By applying the methods of the calculus of variations, show that if there is a Lagrangian of the form $L(q_i, \dot{q}_i, \ddot{q}_i, t)$, and Hamilton's principle holds with the zero variation of both q_i and $\dot{q}_i$ at the end points, then the corresponding Euler–Lagrange equations are

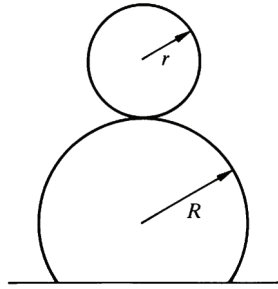
$$\frac{d^2}{dt^2} \left(\frac{\partial L}{\partial \ddot{q}_i} \right) - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) + \frac{\partial L}{\partial q_i} = 0, \quad i = 1, 2, \dots, n.$$

Apply this result to the Lagrangian

$$L = -\frac{m}{2}q\ddot{q} - \frac{k}{2}q^2.$$

Do you recognize the equations of motion?

13. A heavy particle is placed at the top of a vertical hoop. Calculate the reaction of the hoop on the particle by means of the Lagrange's undetermined multipliers and Lagrange's equations. Find the height at which the particle falls off.
14. A uniform hoop of mass m and radius r rolls without slipping on a fixed cylinder of radius R as shown in the figure. The only external force is that of gravity. If the smaller cylinder starts rolling from rest on top of the bigger cylinder, use the method of Lagrange multipliers to find the point at which the hoop falls off the cylinder.



15. A form of the Wheatstone impedance bridge has, in addition to the usual four resistances, an inductance in one arm and a capacitance in the opposite arm. Set up L and $\mathcal{F}$ for the unbalanced bridge, with the charges in the elements as coordinates. Using the Kirchhoff junction conditions as constraints on the currents, obtain the Lagrange equations of motion, and show that eliminating the λ 's reduces these to the usual network equations.
16. In certain situations, particularly one-dimensional systems, it is possible to incorporate frictional effects without introducing the dissipation function. As an example, find the equations of motion for the Lagrangian

$$L = e^{\gamma t} \left(\frac{m\dot{q}^2}{2} - \frac{kq^2}{2} \right).$$

How would you describe the system? Are there any constants of motion? Suppose a point transformation is made of the form

$$s = e^{\gamma t/2} q.$$

What is the effective Lagrangian in terms of s ? Find the equation of motion for s . What do these results say about the conserved quantities for the system?

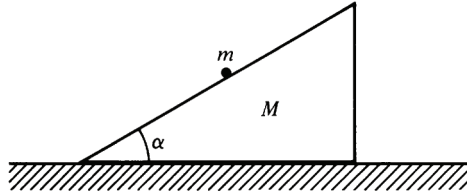
17. It sometimes occurs that the generalized coordinates appear separately in the kinetic energy and the potential energy in such a manner that T and V may be written in the form

$$T = \sum_i f_i(q_i) \dot{q}_i^2 \quad \text{and} \quad V = \sum_i V_i(q_i).$$

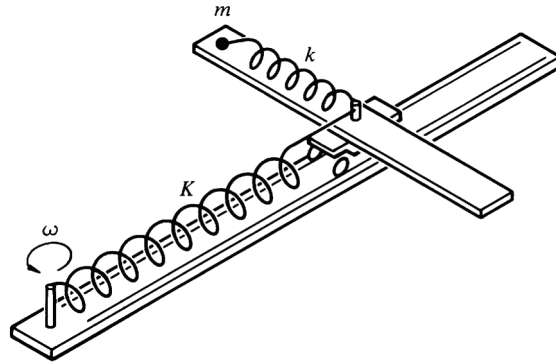
Show that Lagrange's equations then separate, and that the problem can always be reduced to quadratures.

18. A point mass is constrained to move on a massless hoop of radius a fixed in a vertical plane that rotates about its vertical symmetry axis with constant angular speed ω . Obtain the Lagrange equations of motion assuming the only external forces arise from gravity. What are the constants of motion? Show that if ω is greater than a critical value ω_0 , there can be a solution in which the particle remains stationary on the hoop at a point other than at the bottom, but that if $\omega < \omega_0$, the only stationary point for the particle is at the bottom of the hoop. What is the value of ω_0 ?
19. A particle moves without friction in a conservative field of force produced by various mass distributions. In each instance, the force generated by a volume element of the distribution is derived from a potential that is proportional to the mass of the volume element and is a function only of the scalar distance from the volume element. For the following fixed, homogeneous mass distributions, state the conserved quantities in the motion of the particle:
- (a) The mass is uniformly distributed in the plane $z = 0$.
 - (b) The mass is uniformly distributed in the half-plane $z = 0, y > 0$.
 - (c) The mass is uniformly distributed in a circular cylinder of infinite length, with axis along the z axis.
 - (d) The mass is uniformly distributed in a circular cylinder of finite length, with axis along the z axis.
 - (e) The mass is uniformly distributed in a right cylinder of elliptical cross section and infinite length, with axis along the z axis.
 - (f) The mass is uniformly distributed in a dumbbell whose axis is oriented along the z axis.
 - (g) The mass is in the form of a uniform wire wound in the geometry of an infinite helical solenoid, with axis along the z axis.
20. A particle of mass m slides without friction on a wedge of angle α and mass M that can move without friction on a smooth horizontal surface, as shown in the figure. Treating the constraint of the particle on the wedge by the method of Lagrange multipliers, find the equations of motion for the particle and wedge. Also obtain an expression for the forces of constraint. Calculate the work done in time t by the forces of constraint acting on the particle and on the wedge. What are the constants of motion for the system? Contrast the results you have found with the situation when the wedge is

fixed. [Suggestion: For the particle you may either use a Cartesian coordinate system with y vertical, or one with y normal to the wedge or, even more instructively, do it in both systems.]



21. A carriage runs along rails on a rigid beam, as shown in the figure below. The carriage is attached to one end of a spring of equilibrium length r_0 and force constant k , whose other end is fixed on the beam. On the carriage, another set of rails is perpendicular to the first along which a particle of mass m moves, held by a spring fixed on the beam, of force constant k and zero equilibrium length. Beam, rails, springs, and carriage are assumed to have zero mass. The whole system is forced to move in a plane about the point of attachment of the first spring, with a constant angular speed ω . The length of the second spring is at all times considered small compared to r_0 .
- What is the energy of the system? Is it conserved?
 - Using generalized coordinates in the laboratory system, what is the Jacobi integral for the system? Is it conserved?
 - In terms of the generalized coordinates relative to a system rotating with the angular speed ω , what is the Lagrangian? What is the Jacobi integral? Is it conserved? Discuss the relationship between the two Jacobi integrals.



22. Suppose a particle moves in space subject to a conservative potential $V(\mathbf{r})$ but is constrained to always move on a surface whose equation is $\sigma(\mathbf{r}, t) = 0$. (The explicit dependence on t indicates that the surface may be moving.) The instantaneous force of constraint is taken as always perpendicular to the surface. Show analytically that the energy of the particle is not conserved if the surface moves in time. What physically is the reason for nonconservation of the energy under this circumstance?
23. Consider two particles of masses m_1 and m_2 . Let m_1 be confined to move on a circle of radius a in the $z = 0$ plane, centered at $x = y = 0$. Let m_2 be confined to move on a circle of radius b in the $z = c$ plane, centered at $x = y = 0$. A light (massless) spring of spring constant k is attached between the two particles.

- (a) Find the Lagrangian for the system.
- (b) Solve the problem using Lagrange multipliers and give a physical interpretation for each multiplier.
24. The one-dimensional harmonic oscillator has the Lagrangian $L = m\dot{x}^2/2 - kx^2/2$. Suppose you did not know the solution to the motion, but realized that the motion must be periodic and therefore could be described by a Fourier series of the form

$$x(t) = \sum_{j=0} a_j \cos j\omega t,$$

(taking $t = 0$ at a turning point) where ω is the (unknown) angular frequency of the motion. This representation for $x(t)$ defines a many-parameter path for the system point in configuration space. Consider the action integral I for two points, t_1 and t_2 separated by the period $T = 2\pi/\omega$. Show that with this form for the system path, I is an extremum for nonvanishing x only if $a_j = 0$, for $j \neq 1$, and only if $\omega^2 = k/m$.

25. A disk of radius R rolls without slipping inside the stationary parabola $y = ax^2$. Find the equations of constraint. What condition allows the disk to roll so that it touches the parabola at one and only one point independent of its position?
26. A particle of mass m is suspended by a massless spring of length L . It hangs, without initial motion, in a gravitational field of strength g . It is struck by an impulsive horizontal blow, which introduces an angular velocity ω . If ω is sufficiently small, it is obvious that the mass moves as a simple pendulum. If ω is sufficiently large, the mass will rotate about the support. Use a Lagrange multiplier to determine the conditions under which the string becomes slack at some point in the motion.
27. (a) Show that the constraint Eq. (2.29) is truly nonholonomic by showing that it cannot be integrated to a homonomic form.
- (b) Show that the corresponding constraint forces do no virtual work.
- (c) Find one or more solutions to (2.29)–(2.30) for $V = 0$ and show that they conserve energy.